INTERVIEW PREPARATION GUIDE · A TO Z

Marketing Incrementality & Uplift Optimization

Everything you need to explain, defend and demo this project: the business problem, data, tech stack, every model and setting, the statistics behind each number, the results (including the negative ones), and 33 interview questions with model answers.

64,000
randomized customers, 3 arms

9
targeting models compared

+$6.3k
profit / 100k vs propensity targeting (Women's, $3k budget)

54
automated tests

Source of every number: notebooks 01–06 and `reports/results.json` in the project repository. Dataset: Kevin Hillstrom, MineThatData E-Mail Analytics Challenge (2008).

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1 The 60-second pitch

Say this (adapt the wording to your own voice)

"A retailer can send a Men's e-mail, a Women's e-mail, or nothing. The usual approach targets people who are *likely to buy*. Profit, though, comes from people who buy *because of* the e-mail. I used Kevin Hillstrom's randomized experiment, 64,000 customers split into three arms, to estimate those causal effects and turn them into a budget-constrained targeting policy.

First, I verified the randomization and estimated average effects. The Men's e-mail adds about 77 cents of spend per customer and the Women's about 42 cents, both clearly significant. Then I compared a conventional purchase-propensity model with uplift models: T-, S-, X- and DR-learners and an EconML causal forest. I used 5-fold cross-fitting and doubly robust AIPW evaluation, so no model ever graded its own predictions.

The answer was nuanced. For the Women's e-mail, where effects really vary across customers, a regularized causal forest earned about $6,300 more per 100k customers than propensity targeting under a $3k budget, and that gap is statistically significant. For the Men's e-mail there was no learnable heterogeneity, so the best policy is simply to mail everyone.

The biggest lesson was methodological. My pre-registered selection metric, AUGC averaged across campaigns, chose the wrong model. A metric matched to the actual decision, budgeted profit plus a calibration check, would have chosen correctly using development data alone. I also built a Streamlit dashboard that re-optimizes the policy for any margin, cost and budget without ever retraining a model."

Three things the interviewer should remember

1. **Causal, not predictive.** The target is the treatment effect $\tau(x)$, not P(buy). Every evaluation uses the randomized control group.
2. **Rigour.** The project uses known propensities, AIPW, cross-fitting, a test set touched once, paired confidence intervals, Holm correction, an exact MILP check and 54 unit tests.
3. **Honesty and judgement.** Two hypotheses failed, and the project reports why. Uplift only pays where heterogeneity exists, and the selection metric must match the decision.

2 Business problem & research questions

Marketing teams usually score customers by **propensity to buy** and contact the top of the list. That wastes budget on two groups:

Customer type	Buys if e-mailed?	Buys if not e-mailed?	Value of e-mailing
Persuadables	Yes	No	Positive: the only group worth paying for
Sure things	Yes	Yes	Zero (wasted cost). Propensity models rank these at the top
Lost causes	No	No	Zero (wasted cost)
Sleeping dogs	No	Yes	Negative: contact backfires

Uplift modelling estimates the *difference* the e-mail makes for each customer, so the budget goes to persuadables.

Research questions and verdicts

#	Question	Verdict
RQ1	What is the average effect of each campaign?	Both positive and significant (Men's +$0.77, Women's +$0.42 spend per customer)
RQ2	Does uplift targeting beat propensity targeting?	**Only where heterogeneity exists:** yes for Women's, no for Men's
RQ3	How reliably can heterogeneous effects be estimated?	Weakly for spend (≈90% of individual intervals include 0), well for visits
RQ4	Which policy maximizes profit under a budget?	No cap: Men's to everyone. With a cap: causal-forest-targeted Women's ≈ propensity-targeted Men's
RQ5	How sensitive are the conclusions?	Very sensitive to sample size and to the model-selection metric; less to the economics

3 Core concepts in plain words

Potential outcomes

Every customer has two possible outcomes: Y(1) if e-mailed and Y(0) if not. Only one is ever observed. That is the **fundamental problem of causal inference**, and it is why individual effects can never be "checked" directly.

> $\mathrm{ATE} = \mathrm{E}[\,\mathrm{Y}(1) - \mathrm{Y}(0)\,] \qquad \mathrm{CATE}: \tau(\mathrm{x}) = \mathrm{E}[\,\mathrm{Y}(1) - \mathrm{Y}(0) \mid \mathrm{X} = \mathrm{x}\,]$
>
> ATE = average treatment effect over everyone. CATE = conditional average effect for customers with features x. "Uplift" is the CATE.

Why randomization matters

Because the e-mail was assigned by coin-flip, the three groups are statistically identical before the e-mail. Any later difference is caused by the e-mail, so the difference in group means is an unbiased ATE estimate. Randomization also means the **assignment probability e = P(e-mailed) is known (≈ 1/3 per arm)**. Several estimators rely on that.

From effect to decision

> $\text{expected incremental profit of e-mailing customer } \mathrm{i} = \mathrm{m} \cdot \hat{\tau}(\mathrm{x}_\mathrm{i}) - \mathrm{c}$
>
> m = contribution margin (35% of incremental spend, assumed); c = cost per e-mail ($0.10, assumed). E-mail only when this is positive, and only while budget remains.

Two meanings of "propensity" (do not mix them up in an interview)

Term	Meaning	Role here
Purchase propensity	P(buy | X), a customer's predicted chance of buying	The conventional targeting baseline
Propensity score / assignment probability e(x)	P(treated | X)	Known by design (≈1/3); used in IPW/AIPW evaluation

4 The dataset

Kevin Hillstrom, MineThatData E-Mail Analytics and Data Mining Challenge (2008). The population is 64,000 customers who bought in the previous 12 months. They were randomized into three equal groups, and outcomes were tracked for two weeks.

Column	Role	Meaning
recency	feature	Months since last purchase (1–12)
history	feature	Dollars spent in the past year ($29.99–$3,345.93)
history_segment	(dropped)	Binned version of history, so redundant
mens / womens	feature	Bought men's / women's merchandise last year (0/1)
zip_code	feature	Urban / Surburban (sic) / Rural
newbie	feature	New customer in the last 12 months (0/1)
channel	feature	Phone / Web / Multichannel
segment	**treatment**	Mens E-Mail (21,307) / Womens E-Mail (21,387) / No E-Mail (21,306)
visit	outcome	Visited the site within 2 weeks (14.7% overall)
conversion	outcome	Purchased within 2 weeks (0.90% overall, 578 buyers)
spend	**primary outcome**	Dollars spent within 2 weeks ($0 for non-buyers, up to $499)

Engineered features (12 columns)

`recency, history, log_history, mens, womens, newbie, zip_Rural, zip_Surburban, zip_Urban, channel_Multichannel, channel_Phone, channel_Web`. One-hot levels are fixed lists, so every subset of the data produces identical columns.

Data quirks worth mentioning

- **Extreme zero-inflation:** 99.1% of customers spend $0. Spend has SD ≈ $15 against a mean ≈ $1. This single fact explains most of the statistical difficulty.
- **Duplicates (10.3%, 6,562 rows)** are *kept*. There is no customer ID, the features are coarse, and 94% of duplicates are non-visitors with identical profiles. Removing them would distort the randomized sample.
- **Logical checks pass:** no spend without a conversion, and no conversion without a visit.

5 Tech stack & why each tool

Layer	Tool (version)	Used for	Why this choice
Language	Python 3.11.9	Everything	EconML had no Python 3.14 wheels at build time; 3.11 is stable for the whole stack
Data	pandas 3.0.6, NumPy 2.2.6	Loading, features, vectorised metrics	Standard; NumPy pinned <2.3 for EconML compatibility
Statistics	SciPy 1.17.1, statsmodels 0.15.0	Welch tests, chi-square SRM, OLS with HC2 errors (Lin adjustment, BLP), Wald tests, MILP (HiGHS)	Robust standard errors and formula-level control
ML	scikit-learn 1.9.1: RandomForestRegressor (**baseline**), HistGradientBoostingRegressor (**efficiency benchmark**, selected)	Base learners for the response model and every meta-learner, used directly	RF: bagging, variance reduction. HistGB: fast histogram boosting with early stopping; won the NB07 study
ML	XGBoost 3.2.0 (`XGBRegressor` , scikit-learn API) (**primary candidate**)	`base="xgb"` in every meta-learner	Industry-standard boosting; tested head-to-head in NB07 and did not beat the baseline
Causal ML	EconML 0.17.0 (`CausalForestDML`)	Causal forest with honest splitting and pointwise CIs	Reference implementation of generalized random forests with double ML
Optimization	SciPy `milp` (HiGHS)	Exact check of the greedy allocation	Proves the greedy rule is optimal on real data
Visuals	Matplotlib 3.11.2 (notebooks), Plotly 7.1.0 (app)	Static report figures; interactive charts with hover	One shared style module, so every chart uses the same colours per model
App	Streamlit 1.66.0	Decision dashboard	Python-only UI, caching, easy deploy (Streamlit Community Cloud)
Notebooks	Jupyter, nbconvert, nbformat	6 executed notebooks	`nbconvert --execute` reproduces all outputs from scratch
Testing	pytest 9.1.1; `streamlit.testing.AppTest`	54 unit tests; headless app test	Data contract, estimator correctness, budget constraint, optimality
Version control	Git, `.gitignore`	Code, notebooks, reports	Raw data not redistributed; only small derived files for the dashboard are committed

What was deliberately NOT used, and why

CausalML and scikit-uplift were not needed, because the meta-learners are implemented in about 150 transparent lines, which makes them easier to explain in an interview. No database or API was used: the brief scoped this as an offline research prototype. No deep learning: 12 tabular features and a rare outcome favour regularized trees.

6 Architecture & repository map

Pipeline

1 · Ingest
Load CSV, validate data contract, build 12 pre-treatment features

2 · Experiment
SRM, balance, ATEs, Holm, MDE, segment effects

3 · Split
80% dev (5 folds) / 20% test, stratified by arm × conversion

4 · Model
Propensity + 8 CATE models, cross-fitted on dev

5 · Evaluate
AIPW policy values, gain curves, BLP, bootstrap

6 · Decide
Budgeted allocation, sensitivity, dashboard

Layers and leakage controls

Raw data	Read-only CSV. `data.validate()` fails loudly on any contract violation.
Features	Pre-treatment only. A unit test asserts that visit/conversion/spend/segment are never features.
Development (80%)	5-fold cross-fitting: every customer is scored by models that never saw them. All model selection happens here.
Test (20%)	Touched once, in notebook 04, after `model_selection.json` froze the pipeline.
Dashboard	Receives scores and AIPW values only, never raw outcomes or treatment labels, so it cannot retrain or leak.

Repository map

```
src/config.py         frozen config: arms, outcome, seeds, split fractions, economics
src/data.py           load + validate + features + stratified splits / folds
src/inference.py      ATE estimators, balance (SMD), SRM, MDE, Holm, segment effects
src/models.py         ResponseModel, S/T/X/DR-learners, CalibratedPropensity, CausalForest
src/evaluation.py     AIPW/IPW scores, policy gain, gain & Qini curves, BLP calibration
src/optimization.py   net value, budgeted multi-arm allocation, MILP, paired comparison
src/pipeline.py       cross_fit(), fit_and_score(), evaluation_scores(), summaries
src/scenario.py       policy scenarios shared by notebooks 04/05 and the dashboard
src/benchmark.py      base-learner study: grids, nested CV, timing (notebook 07)
src/plotting.py       one chart style; colour follows the model, never its rank
notebooks/01–07       the six phases + the base-learner study, executed with outputs
app/dashboard.py      Streamlit decision tool
tests/                54 pytest tests
reports/              research_report.md, figures/, results.json, reproducibility.json
artifacts/            generated CSV/JSON passed between notebooks
```

Design rule: notebooks *explain* and call `src/`; they never duplicate the logic. The dashboard and notebook 04 call the same `scenario.run_scenario()`, so their numbers match to the cent, and notebook 05 asserts it.

7 Module-by-module code walkthrough

config.py: one source of truth

Treatment labels, primary campaign (Women's), primary outcome (spend), SEED = 42, robustness seeds [7, 11, 23, 101, 2024], SPLIT_FRACTIONS {dev 0.8, test 0.2}, N_FOLDS = 5, MARGIN = 0.35, COST_PER_CONTACT = 0.10, POPULATION_SIZE = 100,000, DEFAULT_BUDGET = $3,000 (enough to e-mail 30%).

data.py

- `validate()`: required columns, no nulls, exact arm labels, binary flags, recency 1..12, non-negative money, consistency rules. Raises `DataValidationError`.
- `build_features()`: 12 numeric columns with fixed one-hot levels; asserts no outcome leakage.
- `assign_splits()`: stratified by *arm* × *conversion*, so the rare buyers spread evenly. `assign_folds()` reuses it for 5 equal folds.
- `arm_shares()`: empirical arm proportions, used as the known propensities.

inference.py

- `diff_in_means` (Welch interval), `bootstrap_ate` (resample within arm), `permutation_test` (randomization inference).
- `lin_regression_ate`: Y ~ T + Xc + T·Xc with HC2 errors. Collinear one-hot columns are dropped greedily by rank.
- `balance_table` (SMD), `srm_test` (chi-square), `minimum_detectable_effect`, `holm`, `segment_effects`.

models.py: shared interface `fit(X, t, y)` / `predict(X) → τ̂`

Every learner accepts `base="hgb" | "rf" | "xgb"` through its params, e.g. `make_model("t_learner", base="rf")`. The shared knobs `min_samples_leaf` and `max_leaf_nodes` are translated per family, so "regularized" means the same thing for each.

Class	Algorithm	Key settings
base_regressor(base=...)	**hgb**: HistGradientBoosting (default)	lr 0.05, max_iter 300, max_leaf_nodes 15, min_samples_leaf 200, L2 1.0, early stopping (15% validation, patience 20)
	rf: RandomForest	300 trees, min_samples_leaf 200, max_leaf_nodes 120, max_features 0.5, max_samples 0.5
	xgb: XGBoost	300 rounds, lr 0.05, lossguide max_leaves 15, min_child_weight 200, subsample 0.8, colsample 0.8, λ 1.0 (no early stopping)
ResponseModel	HistGB classifier on *e-mailed customers only*, target = conversion	Same as base
SLearner	One model on [X, T]; $\hat{\tau} = \mu(x,1) - \mu(x,0)$	Base
TLearner	μ_0 on control, μ_1 on treated; $\hat{\tau} = \mu_1 - \mu_0$	Base

XLearner	Imputed effects + effect models + propensity weighting	Default, or **regularized**: min_samples_leaf 600, max_leaf_nodes 8
DRLearner	Regress 3-fold cross-fitted AIPW pseudo-outcomes on X	Final model min_samples_leaf 400
CalibratedPropensity	$\hat{\tau} = a + b \cdot p(x)$, with (a, b) fitted by OLS of out-of-fold pseudo-outcomes on out-of-fold p(x)	Hypothesis; rejected
CausalForest	EconML CausalForestDML: model_y = HistGB, model_t = DummyClassifier(prior), honest, inference on	400 trees, min_samples_leaf 50 or **200**, max_samples 0.45, cv 3

evaluation.py

- `aipw_scores()`: multi-arm Γ, with outcome models cross-fitted *within* the evaluation partition.
- `top_k_mask`, `policy_gain`, `gain_at_coverage`, `gain_curve`, `area_above_random` (AUGC).
- `qini_curve`, `qini_coefficient`, `bootstrap_qini`; `uplift_by_bin`; `calibration_slope` (BLP).

optimization.py

- `net_value` = $m \cdot \hat{\tau} - c$. `allocate()` is the greedy multi-arm budgeted allocation: each customer gets their best arm, positive net only, then customers are admitted by net per dollar until the budget binds.
- `solve_milp()`: exact 0/1 solution with sparse constraints (HiGHS). `evaluate_allocation`, `profit_contributions`, `compare_allocations` (paired difference).

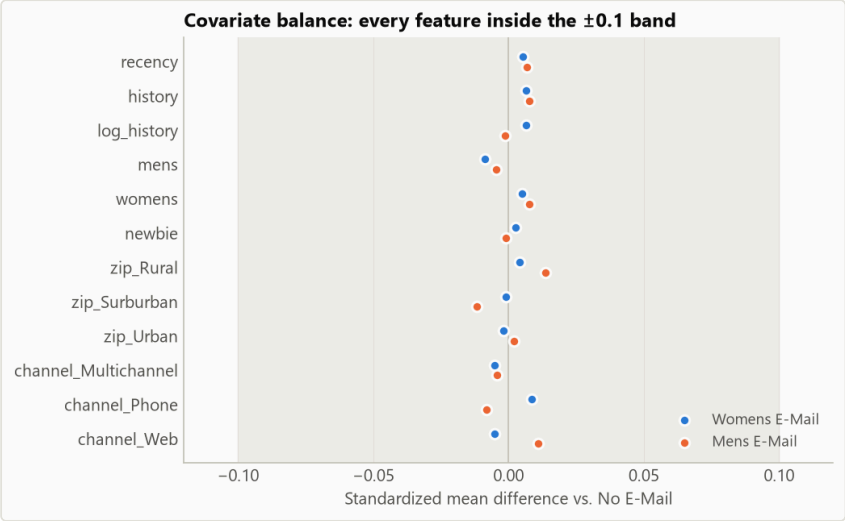
pipeline.py & scenario.py

`fit_and_score()` trains on (campaign arm + control) rows and scores all three arms. `cross_fit()` produces out-of-fold scores for every development customer, and `cross_fit_gamma()` produces matching AIPW scores computed within each fold. `scenario.run_scenario()` evaluates every policy for given economics and is the dashboard's engine.

8 Phase 1 · Data understanding & experimental analysis (NB01)

Was randomization valid?

Check	Method	Result
Sample-ratio mismatch	Chi-square of arm counts vs 1/3 each	χ^2 = 0.20, **p = 0.90**
Covariate balance	Standardized mean difference, 12 features × 2 arms	**max |SMD| = 0.014** (threshold 0.1)
Joint balance	Logistic regression predicting assignment, 5-fold AUC	**0.496 / 0.499** (0.5 = no information)

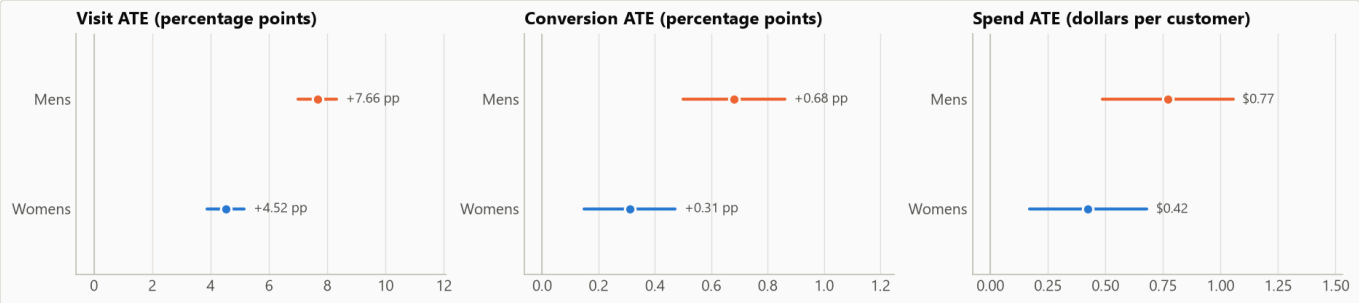


Every feature is inside the ±0.1 SMD band for both campaigns.

Average treatment effects (four estimators agree)

Campaign	Visit	Conversion	Spend / customer (95% CI)	Relative spend lift	Holm p
Men's	+7.66 pp	+0.68 pp	**$0.770** (0.485–1.055)	+118%	3.5e-07
Women's	+4.52 pp	+0.31 pp	**$0.424** (0.169–0.680)	+65%	1.1e-03

- **Estimators:** Welch difference in means (primary), within-arm bootstrap (2,000), permutation test (2,000), and Lin regression adjustment. All agree.
- **Lin adjustment** reduced variance by at most 2.9% (visit) and ~0.1–0.2% for spend, because the features explain little of the outcome.
- **Head-to-head:** Men's beats Women's on spend by $0.35 (p = 0.03).
- **Mechanism:** spend per buyer is similar across arms ($114, $114, $122), so the lift comes from *more buyers*, not bigger baskets. This comparison is descriptive only, because it conditions on a post-treatment variable.



ATE with 95% CI for each outcome.

Power: the warning that shaped the whole project

The minimum detectable effect (80% power, $\alpha = 0.05$) for spend on a 20% test split is **$0.82 (Women's)**, about **twice** the Women's ATE. A single test split therefore cannot rank policies reliably. This is why the project uses AIPW, cross-fitting and, finally, a pooled evaluation over all 64,000 customers.

Is there heterogeneity worth modelling? Check before you model

Joint Wald test (all T × X interactions = 0)	Visit p	Spend p
Women's e-mail	≈ 0 (χ^2 = 106.5)	**0.014**
Men's e-mail	0.0013	**0.57**

> **Interview gold**
>
> This test *predicted* the final result. The Women's spend effect is heterogeneous and the Men's is not, and uplift targeting later worked for Women's and failed for Men's. Run a cheap heterogeneity test before investing in uplift models.

Segment effects (Holm-adjusted within families) show the Women's e-mail works for new customers, women's-merchandise buyers and multichannel buyers. It barely moves existing customers or phone-only buyers. The Men's e-mail lifts spend almost uniformly.

9 Evaluation framework: AIPW & cross-fitting

Individual effects are never observed, so a policy cannot be scored by "accuracy". Instead, each customer gets a doubly robust score whose expectation is their CATE.

> $$\Gamma_{ik} = \hat{\mu}_k(X_i) - \hat{\mu}_0(X_i) + 1[A_i = k]\cdot(Y_i - \hat{\mu}_k(X_i)) \,/\, e_k - 1[A_i = 0]\cdot(Y_i - \hat{\mu}_0(X_i)) \,/\, e_0$$
>
> $\hat{\mu}_k$ = predicted outcome in arm k; e_k = known assignment probability (≈ 1/3). The first two terms are a model-based effect; the last two are inverse-propensity-weighted corrections using the observed outcome.

Why it is trustworthy

- **Unbiased even if $\hat{\mu}$ is wrong.** Because e_k is known, $E[\Gamma_{ik} \mid X_i] = \tau_k(X_i)$. A poor outcome model only increases variance. This is the "doubly robust" property in its strongest form.
- **Lower variance than IPW.** The reduction was ~15–17% for visit, but only ~0.3% for spend, because features barely predict spend. AIPW is never worse, so it is kept.
- **Independent nuisance models.** $\hat{\mu}$ is cross-fitted *inside each evaluation fold*, independent of every candidate model, so no model grades its own homework.

Policy value and metrics

> $$\text{incremental value of e-mailing set S with campaign k} = (1/n)\,\Sigma_{i\in S}\,\Gamma_{ik} \quad \text{(CLT interval)}$$
>
> Profit of a policy = $(1/n)\,\Sigma_{i\in S}\,(m\cdot\Gamma_{ik} - c)$. Two policies on the same customers are compared with a **paired** difference, which has a much tighter interval.

Metric	Definition	Reads as
Gain curve	Cumulative AIPW gain from e-mailing the top φ by score	Incremental $ per population customer vs share e-mailed
AUGC	Area between the gain curve and the random line	$ per customer above random, averaged over coverage (0 = random)
Qini curve / coefficient	$Y_T(\varphi) - Y_C(\varphi)\cdot N_T(\varphi)/N_C(\varphi)$, area above the diagonal	The classic uplift metric on a two-arm sample (reported for completeness)
gain@30% / vs_random	Gain at 30% coverage; paired difference from random	The decision-relevant point for a 30% budget
BLP slope	OLS slope of Γ on predicted CATE (Chernozhukov et al.)	1 = predictions correctly scaled; 0 = pure noise

Cross-fitting (development set)

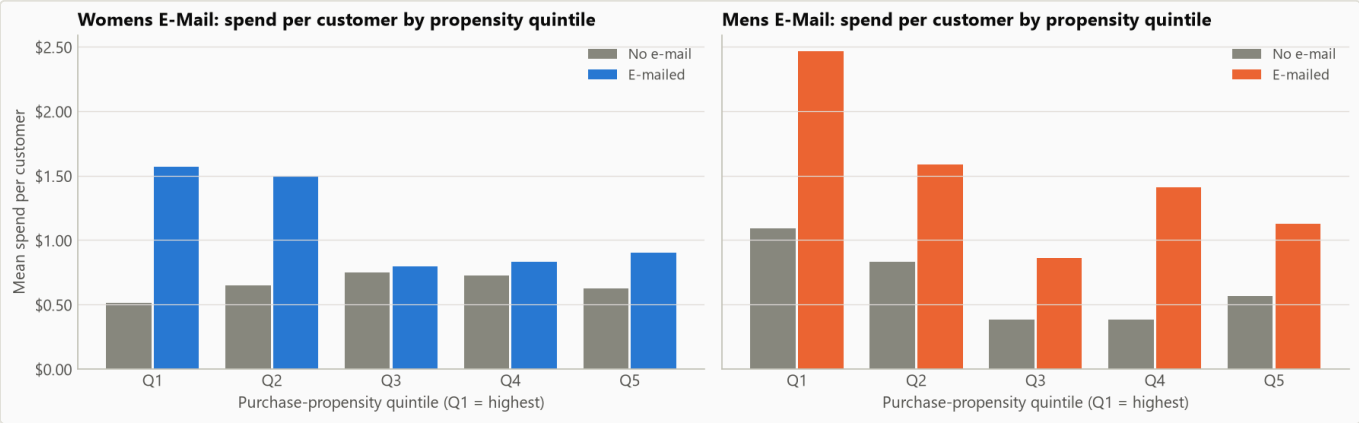
The 80% development set is split into 5 folds. For each fold k, models are trained on the other 4 folds and score fold k, and fold k's Γ is computed using fold k only. All 51,200 development customers then have honest out-of-fold predictions. That gives about half the standard error of a single 20% validation split.

10 Phase 2 · Baseline models (NB02)

Policy	Who gets the e-mail
No campaign	Nobody (reference = $0)
Treat all	Everyone
Random	Random share φ; expected value = φ × ATE (computed analytically)
Purchase propensity	Top φ by P(buy | X, e-mailed)
T-learner	Top φ by $\hat{\mu}_1 - \hat{\mu}_0$

Purchase-propensity model

- Out-of-fold AUC is about 0.57–0.58. That is weak, but typical for a <1% outcome with 8 coarse features, and the model is reasonably calibrated across deciles.
- **The revealing diagnostic is which group it ranks.** Women's model: AUC 0.573 on e-mailed customers vs **0.529 on control**, so it learned *responsiveness*. Men's model: 0.579 e-mailed vs **0.621 control**, so it learned *sure things*.
- The identity behind it: P(buy | X, e-mailed) = P(buy | X, not e-mailed) + τ(X). A model fitted on e-mailed customers mixes baseline demand with uplift, and the mix differs by campaign.



Women's: control spend is flat across propensity quintiles, and the e-mail gap sits at the top (implicit uplift model). Men's: control spend also rises at the top ("sure things").

T-learner

Its predicted effects have SD ≈ $1.10 around a $0.46 mean, and 25% of them are negative. Its AUGC is −0.011 (no better than random) and its BLP slope is −0.09, so the predictions carry no held-out signal. The CATE is a difference of two noisy regressions on a 0.9%-nonzero outcome, and the noise of both models adds up.

> **Phase-2 takeaway**
>
> A strong, low-variance non-causal baseline (propensity) beat the naive causal model (T-learner) for Women's. Advanced models must beat *that*. Also, a ranking cannot decide *how many* customers to e-mail or *which* campaign each should get, because those decisions need effects in dollars.

11 Phase 3 · Advanced uplift models (NB03)

How each learner works

S-learner

One model with T as a feature. Regularization can shrink τ toward 0, because trees may never split on T.

T-learner

Two separate models. Simple, but it differences two noisy fits.

X-learner (Künzel et al. 2019)

1. Fit μ_0 and μ_1.
2. Impute effects: $D_1 = Y - \mu_0(X)$ for treated customers, $D_0 = \mu_1(X) - Y$ for controls.
3. Regress D_1 and D_0 on X to get τ_1 and τ_0.
4. Combine: $\tau = e \cdot \tau_0 + (1-e) \cdot \tau_1$.

Designed for weak signal and imbalanced arms.

DR-learner (Kennedy 2020)

Compute cross-fitted AIPW pseudo-outcomes Γ, then regress Γ on X. The regression target is unbiased because e is known.

Causal forest (Wager & Athey 2018)

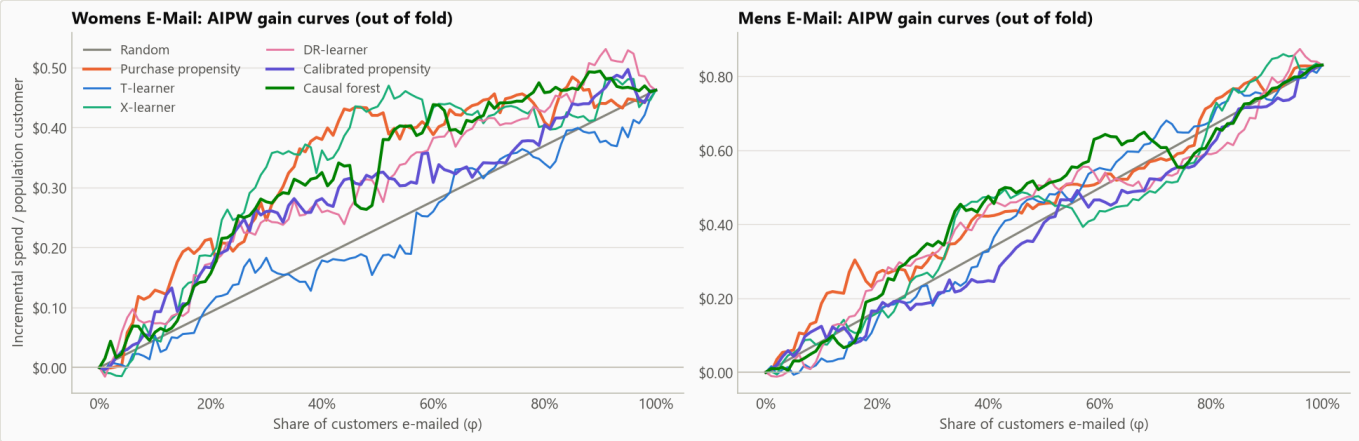
Trees split to maximize *effect* heterogeneity, not outcome fit. **Honesty** means one half-sample chooses the splits and the other estimates the leaf effects, which enables valid **confidence intervals**. The DML variant first residualizes Y and T (double machine learning).

Calibrated propensity (my hypothesis)

$\tau = a + b \cdot p(x)$: the strong Women's propensity ranking plus a 2-parameter DR calibration into dollars.

Out-of-fold results on development data (spend)

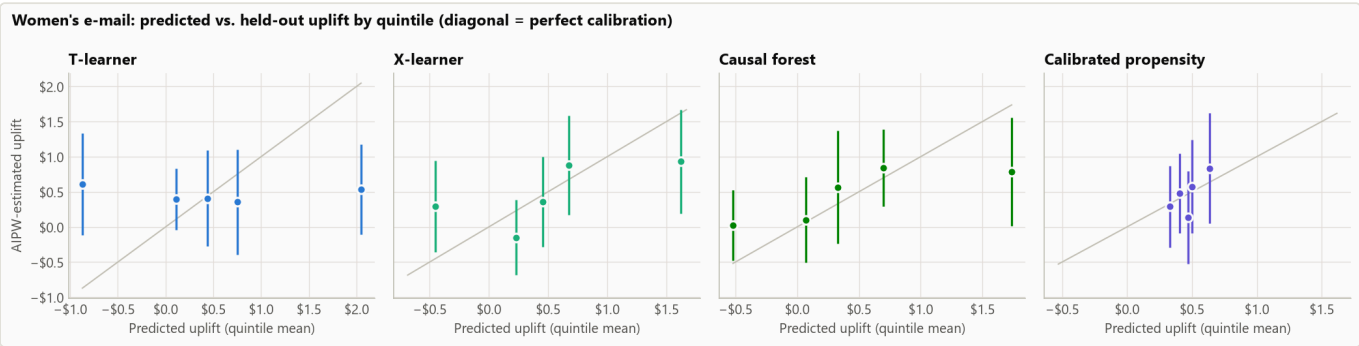
Model	Mean AUGC	Women's AUGC	Men's AUGC	BLP Women's	BLP Men's
Purchase propensity	**+0.076**	+0.105	+0.047	n/a	n/a
X-learner (regularized)	+0.073	+0.138	+0.008	0.69	0.004
Causal forest (leaf 200)	+0.071	+0.108	+0.034	0.79	0.31
Causal forest (leaf 50)	+0.066	+0.087	+0.045	0.34	0.14
X-learner	+0.058	+0.102	+0.015	0.22	0.09
DR-learner	+0.047	+0.072	+0.023	0.27	0.04
Calibrated propensity	+0.013	+0.055	−0.029	1.02	−0.24
S-learner	+0.011	+0.040	−0.018	0.14	−0.15
T-learner	−0.002	−0.011	+0.007	−0.09	−0.04



Out-of-fold AIPW gain curves. Curves above the grey random line add value.

Uncertainty and robustness findings

- **Bootstrap (300 paired replicates):** all AUGC intervals overlap heavily. The selected model vs propensity is −0.003 ± 0.041.
- **Causal-forest CIs:** about 90% of customers' 95% intervals include zero (median width $2.23). Individual effects are not identifiable with confidence, even though rankings still carry signal on average.
- **Regularization beats algorithm choice:** heavier leaves raise the Women's BLP slope from 0.22 to 0.69 (X-learner) and from 0.34 to 0.79 (forest).
- **Seed stability** (rank correlation across 3 seeds): causal forest leaf 200 0.96 (top-30% Jaccard 0.84), causal forest 0.92, X-learner regularized 0.85, propensity 0.77, T-learner 0.67, DR-learner 0.60.
- **Visits as KPI:** T/X/forest models trained on visits reach AUGC 0.0072–0.0076 vs propensity 0.0030 (≈2.5×), and beat random by 5–7 standard errors. Spend is a signal-to-noise problem, not a broken method.
- **Segment calibration:** T-learner segment averages track observed segment effects (r = 0.98), even though its individual ranking is noise. Real heterogeneity lives *between coarse segments*.
- **Calibrated propensity failed:** its (a, b) were unstable across folds. One Women's fold collapsed to near-constant predictions (SD $0.012), and one Men's fold had a negative slope that inverted the ranking (ρ = −1). Rejected.

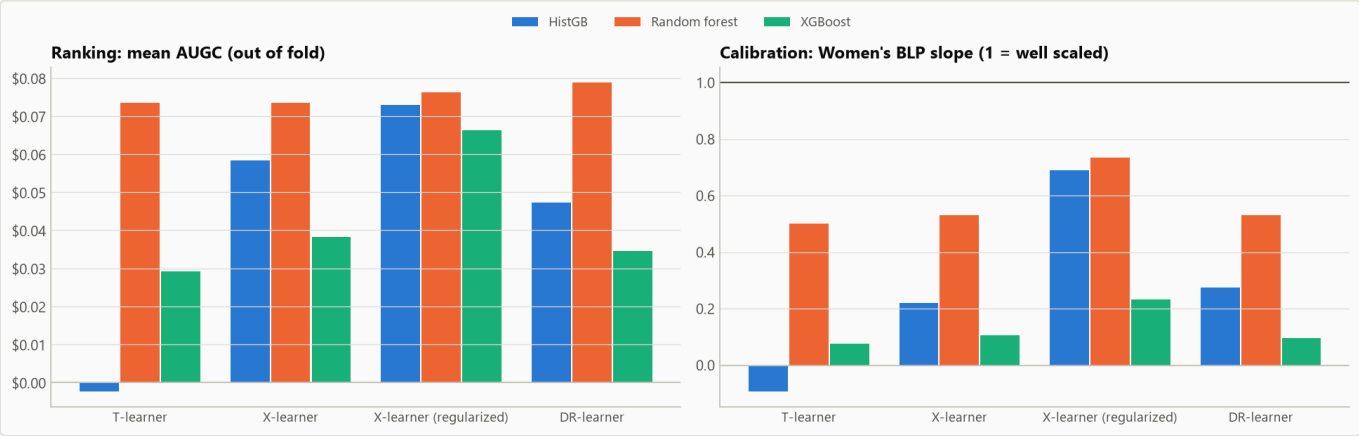


Predicted vs held-out uplift by quintile; the diagonal means perfect calibration.

Base-learner comparison at default settings (NB03 §4c)

As a first look, the T-, X-, regularized X- and DR-learners were re-fitted with all three base learners at **fixed default settings**, on the same 5 folds and the same AIPW scores. This is a development-data sensitivity analysis. The decisive, tuned, held-out comparison is in **section 12**, and it changes the conclusion.

Learner	AUGC HistGB	AUGC RF	AUGC XGB	W BLP HistGB	W BLP RF	W BLP XGB
T-learner	−0.002	**+0.074**	+0.029	−0.09	**0.50**	0.08
X-learner	+0.058	**+0.074**	+0.038	0.22	**0.53**	0.11
X-learner (regularized)	+0.073	**+0.076**	+0.066	0.69	**0.74**	0.23
DR-learner	+0.047	**+0.079**	+0.035	0.27	**0.53**	0.10
Mean of 4	+0.044	**+0.076**	+0.042	0.27	**0.57**	0.13



Random Forest (orange) leads on both ranking and calibration; XGBoost (green) is the weakest calibrated.

- **At default settings, Random Forest looks best:** it rescues the T-learner and gives the only meaningful Men's calibration in the project (RF regularized X-learner, Men's BLP ≈ 0.41). Once each base learner is tuned with an equal budget and judged on held-out policy value (section 12), the three are statistically tied, and HistGB wins on point estimate and speed.
- **Why:** bagging averages many independently fitted trees, which cancels the noise of a rare, heavy-tailed outcome. Boosting fits residuals in sequence and chases a few big purchases, and differencing (T) or imputing (X) amplifies that.
- **Fairness caveat:** XGBoost ran a fixed 300 rounds without early stopping (HistGB has it), which partly explains its weak calibration.
- **For the budgeted decision** (dev profit@30%), the best combinations are within ≈ $0.9k of each other, inside one SE (≈ $3.4k). No base learner is a clear winner for the decision itself.

Pre-registered selection

Rule: the highest mean AUGC among dollar-valued models, with a tie-break that prefers a simpler model within one bootstrap SE. **Selected: X-learner (regularized). Challenger: causal forest (leaf 200).** The decision was frozen in `artifacts/model_selection.json` before the test set was opened.

12 Base learners: Random Forest, XGBoost, HistGradientBoosting

A T-learner is a *recipe*: fit one outcome model per arm and subtract, $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$. The **base learner** is the regression algorithm that plays $\hat{\mu}$. This project tests three, chosen on held-out evidence rather than reputation (notebook 07).

Role	Base learner	Library (used directly)	Family
Baseline	Random Forest	`sklearn.ensemble.RandomForestRegressor`	Bagging
Primary candidate	XGBoost	`xgboost.XGBRegressor` (scikit-learn API)	Gradient boosting (second order)
Efficiency benchmark	HistGradientBoosting	`sklearn.ensemble.HistGradientBoostingRegressor`	Histogram gradient boosting

How each one works

Random Forest: many independent trees, averaged

1. Draw a random sample of rows (here 50%) for each tree.
2. Grow a deep decision tree. At each split, consider only a random subset of features (here 33–67%), and stop when a leaf would hold fewer than `min_samples_leaf` customers.
3. Repeat 300 times and **average** the predictions.

Why it helps: each tree is noisy, but their errors are partly independent, so averaging cancels variance without adding bias. A leaf predicts the *average of real outcomes*, so it cannot extrapolate to extreme values, which acts as built-in shrinkage. **Cost:** it has hundreds of deep trees to build and to walk through at prediction time.

XGBoost: trees added one at a time to fix the previous errors

The model is a sum of shallow trees, $F(x) = \Sigma_m \eta \cdot f_m(x)$. Each new tree is fitted to the gradient g and Hessian h of the loss left by the trees so far. Its leaf weights and split gains have a closed form:

> $\text{leaf weight } w^* = -\Sigma g / (\Sigma h + \lambda) \quad \text{split gain} = \tfrac{1}{2}\left[G_L^2/(H_L+\lambda) + G_R^2/(H_R+\lambda) - (G_L+G_R)^2/(H_L+H_R+\lambda) \right] - \gamma$
>
> For squared loss, g = prediction − y and h = 1, so `min_child_weight` (minimum Σh in a leaf) is simply a minimum leaf size. Regularization comes from λ (L2 on leaf weights), the learning rate η, row and column subsampling, and the number of rounds.

Why it is usually strong: it fits structure step by step, with precise second-order updates. **The risk here:** every round fits what the previous trees missed. On a heavy-tailed outcome, much of what is "missed" is noise from a few large purchases, so without early stopping the number of rounds controls how much noise it learns.

HistGradientBoosting: the same boosting idea, made fast

Before training, each feature is binned into at most 255 histogram buckets, so finding the best split means scanning a few hundred bins instead of every unique value. Trees grow leaf-wise with a maximum leaf count, and **early stopping** holds out 15% of the training rows and stops adding trees when validation loss stops improving for 20 rounds. It is scikit-learn's implementation of the approach popularized by LightGBM.

Side-by-side

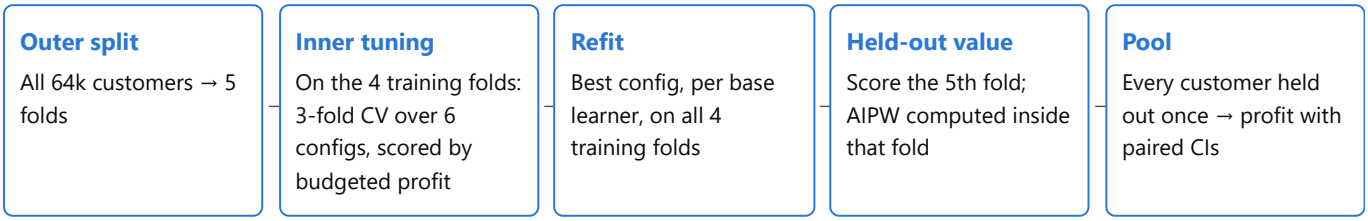
	Random Forest	XGBoost	HistGradientBoosting
Trees built	In parallel, independent, deep	Sequentially, shallow	Sequentially, shallow, on binned features
Main error reduced	Variance	Bias (then variance through regularization)	Bias (variance through early stopping)
Key knobs here	min_samples_leaf, max_features	min_child_weight, n_estimators	min_samples_leaf, max_leaf_nodes
Overfitting guard	Averaging + leaf size	λ, η, subsampling, round count (no early stopping)	Early stopping + leaf size + L2
Strength	Robust and well-calibrated effect sizes	Most expressive	Fastest, tuning-stable
Weakness	Slowest to fit and predict	Fits noise on rare outcomes	Slightly less calibrated than RF

How they are used in the project

```
# any meta-learner, any base learner (src/models.py)
models.make_model("t_learner", base="rf",  min_samples_leaf=200)   # Random Forest
models.make_model("t_learner", base="xgb", min_samples_leaf=200)   # XGBoost
models.make_model("x_learner", base="hgb", min_samples_leaf=600, max_leaf_nodes=8)
```

- **One factory, shared knobs.** `min_samples_leaf` maps to RF's `min_samples_leaf`, XGBoost's `min_child_weight` and HistGB's `min_samples_leaf`. `max_leaf_nodes` maps to `max_leaf_nodes` / `max_leaves`. Any other keyword passes straight through to the library estimator.
- **Defaults:** RF uses 300 trees, max_features 0.5, max_samples 0.5. XGBoost uses 300 rounds, η 0.05, lossguide, subsample 0.8, colsample 0.8, λ 1. HistGB uses η 0.05, max_iter 300, L2 1, and early stopping (default for the whole pipeline).
- **Equal tuning grids** (6 configurations each, `src/benchmark.py`). RF: leaf {100, 200, 400} × max_features {0.33, 0.67}. XGBoost: leaf {100, 200, 400} × rounds {50, 200}. HistGB: leaf {100, 200, 400} × max leaves {7, 15}.

The study design (notebook 07)

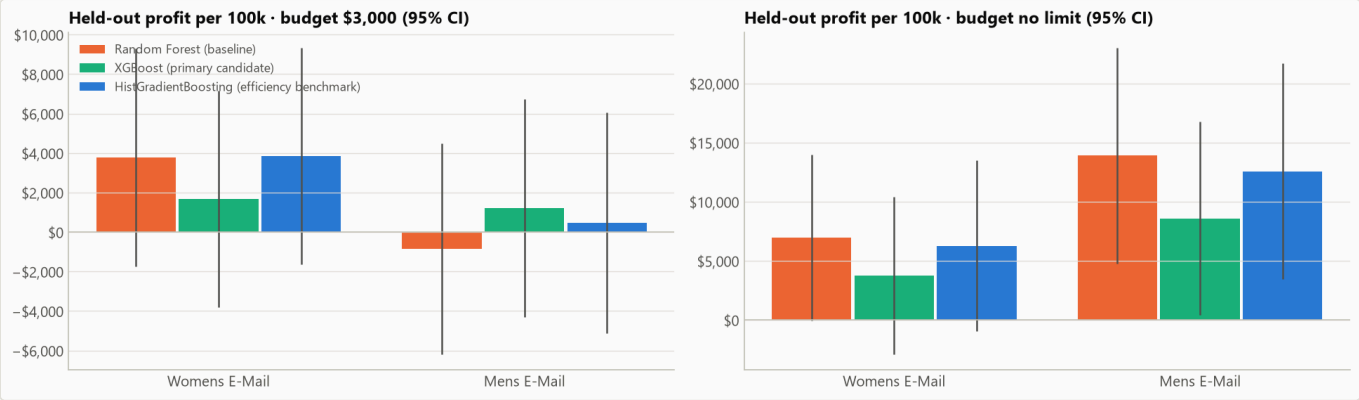


This is **nested cross-validation**. No tuning decision ever sees the customers it is evaluated on, and pooling all 64k gives about half the standard error of a single 20% test split. The winner rule was declared in advance: the highest pooled held-out profit at the $3k budget, averaged over campaigns.

Results

Held-out (nested CV), per 100k customers	Random Forest	XGBoost	HistGB

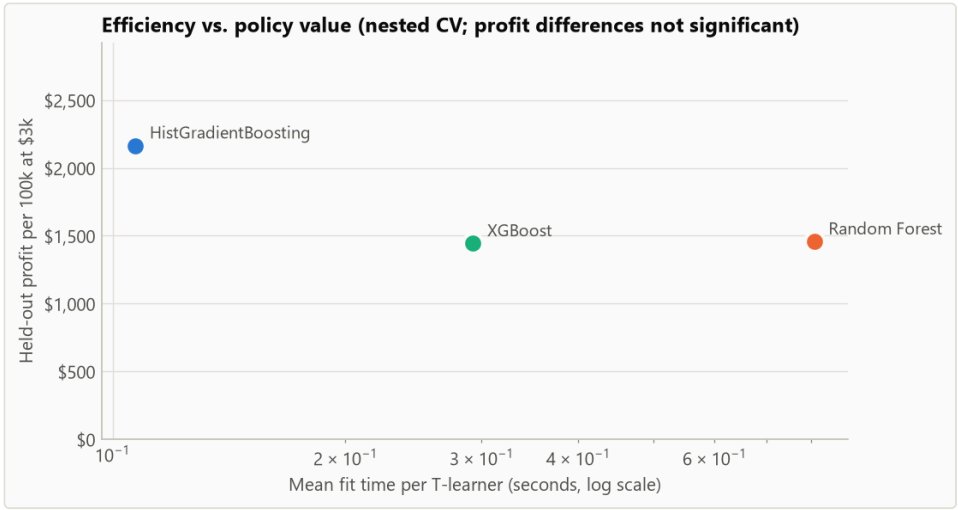
Profit @ $3k, Women's	$3,780	$1,667	$3,854
Profit @ $3k, Men's	−$867	$1,219	$474
Profit @ $3k, mean of campaigns	$1,457	$1,443	**$2,164**
Profit, no budget, Women's	**$6,975**	$3,740	$6,274
Profit, no budget, Men's	**$13,903**	$8,600	$12,584
Women's AUGC (ranking)	**+0.070**	−0.012	+0.052
Women's BLP (calibration)	**0.15**	−0.01	0.05
Paired vs RF @ $3k (95% CI)	—	−$14 (−$3.2k, +$3.1k)	+$707 (−$2.4k, +$3.9k)
Same config in every outer fold?	No	No	Max leaves = 7 always



Held-out (nested CV) profit per 100k with 95% CI. Left: $3k budget. Right: no budget limit (break-even rule).

Efficiency comparison

Per T-learner (two outcome models, ~27k rows)	Random Forest	XGBoost	HistGB
Mean fit time	0.81 s	0.29 s	**0.11 s**
Mean predict time (12.8k rows)	0.176 s	0.015 s	**0.010 s**
Total fit time for the whole study (250 fits)	202 s	73 s	**27 s**



Fit time (log scale) vs held-out profit at $3k. The profit differences are not statistically significant; the speed differences are large.

Verdict

Selected: HistGradientBoosting

- **It is the pre-declared winner:** highest held-out profit at the $3k budget. The edge over RF is a statistical tie.
- **It is the most efficient:** 2.7× faster to fit than XGBoost, 7.5× faster than Random Forest, and far faster to predict.
- **It is the most tuning-stable:** it chose the same leaf count in every outer fold.
- **Exception:** when a policy relies on **break-even thresholds without a budget**, Random Forest's better-calibrated effect sizes earn the most. Use RF there.
- **XGBoost** (the primary candidate) did **not** beat the baseline. Without early stopping its dollar predictions were uncalibrated, and it lost $3–5k per 100k when no budget was set.

Why the "strongest" algorithm did not win

A T-learner subtracts two models of an outcome where 99.1% of values are $0. Whatever noise each model learns survives the subtraction, so the base learner that **fits the least noise** wins. XGBoost's extra expressiveness is a liability unless it is held back by early stopping. Early-stopped HistGB and averaged Random Forests both hold back.

Why not other algorithms?

Algorithm	Why it was not included
LightGBM	HistGradientBoosting is scikit-learn's implementation of the same histogram/leaf-wise idea, so LightGBM would add a dependency for a near-duplicate.
CatBoost	Its main advantage is native categorical handling, but there are only two small categorical fields (zip, channel), one-hot encoded. It is slower to train.
Linear / Lasso / logistic	Cannot capture interactions (e.g. new customer × channel) without manual feature engineering. A linear model *was* used for the heterogeneity Wald test, where interpretability matters.
Neural networks	12 tabular features and a 0.9% outcome: they would overfit, need heavy tuning, and offer no advantage on this kind of data.
k-NN / SVR	Sensitive to feature scaling and the mix of binary and continuous inputs. SVR does not scale well to ~50k rows.
Tweedie / two-part models	A good fit for zero-inflated spend in principle (HistGB supports Poisson loss). Listed as future work rather than mixed into this comparison.

All three chosen learners are tree ensembles because trees handle mixed binary and continuous features without scaling, capture interactions automatically, and are robust to the heavy-tailed history feature.

13 Phase 4 · Policy evaluation & optimization (NB04)

The optimization problem

> $\text{maximize } \Sigma_i \Sigma_k a_{ik} (m \cdot \hat{\tau}_{ik} - c) \quad \text{subject to } \Sigma_i \Sigma_k a_{ik} \cdot c \leq B, \quad \Sigma_k a_{ik} \leq 1, \quad a_{ik} \in \{0,1\}$
>
> Each customer gets at most one campaign; total spend on e-mails ≤ budget B.

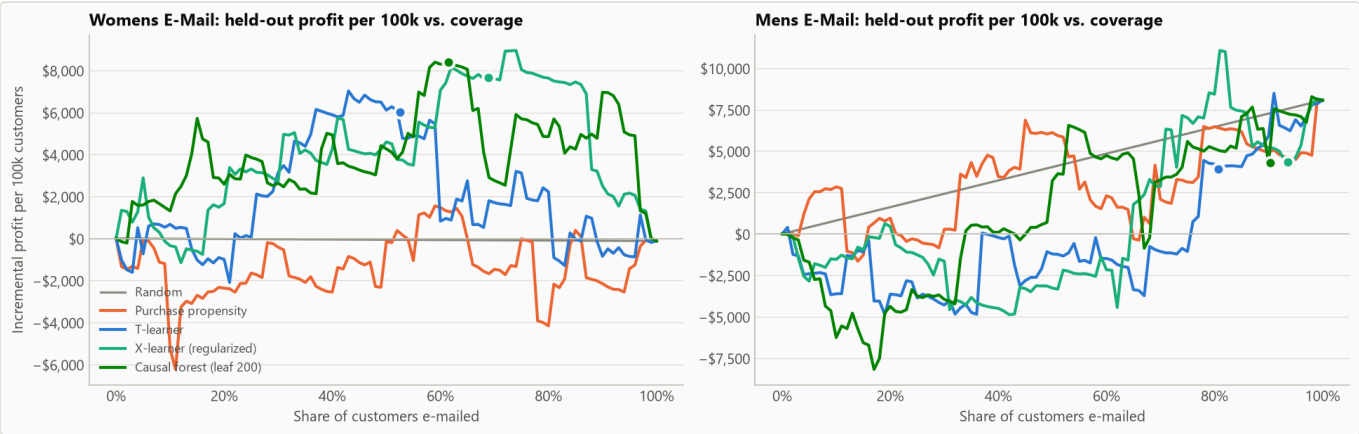
Why greedy is exactly optimal here. Every e-mail costs the same, so the budget constraint is just a cap on the *number* of contacts. The optimum is therefore: (1) pick each customer's best campaign, (2) drop customers with negative net value, (3) take the top-N by net value. Verified on the test set: greedy and MILP objectives are identical (1562.91), and greedy takes 2 ms vs 13.3 s for the MILP. With unequal costs this becomes a knapsack problem, and greedy-by-ratio is only a heuristic.

How each policy spends the budget

- **Random / propensity rankings** fill the whole budget on the campaign with the larger *development* ATE (Men's). A ranking cannot tell whether a contact pays for itself.
- **Uplift policies** contact a customer only if $m \cdot \hat{\tau} > c$, so they may spend less than the budget, and can choose the campaign per customer.

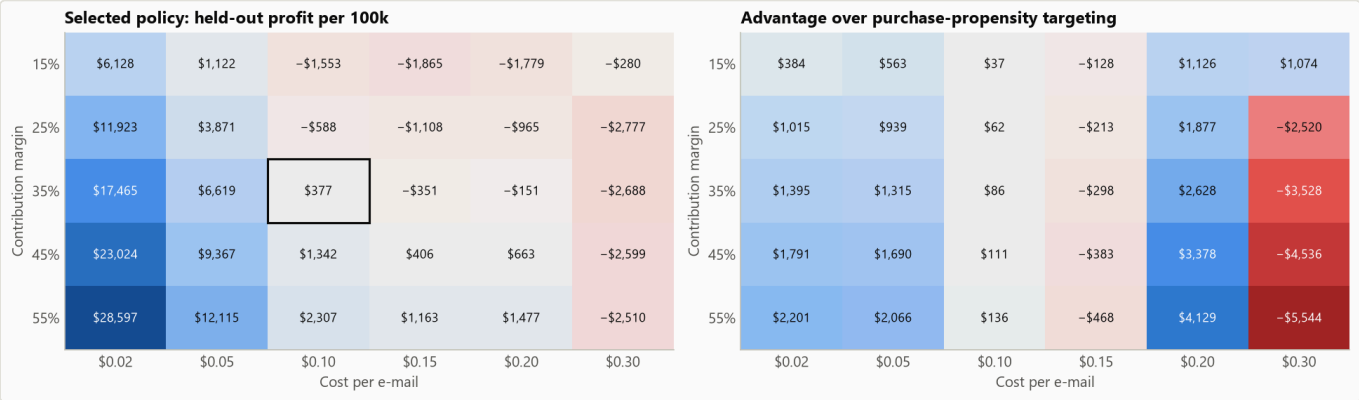
Single test split (12,800 customers)

- **Women's ranking replicates:** AUGC +0.12 (X-learner and forest) vs −0.04 (propensity). BLP slopes are 0.81 and 0.93, so the dollar predictions are well calibrated out of sample.
- **Men's:** every model is at or below random (BLP < 0).
- **No policy comparison is significant:** intervals span about ±$10k per 100k. This test split was also "unlucky": its ATEs ($0.28 / $0.52) fall below the development estimates ($0.46 / $0.83).
- **Break-even rule** (no budget): uplift models choose to e-mail 53–69% for Women's, earning $6–8k per 100k, vs about $0 for treat-all.
- **Checks:** IPW agrees with AIPW within noise, and the budget is never exceeded.



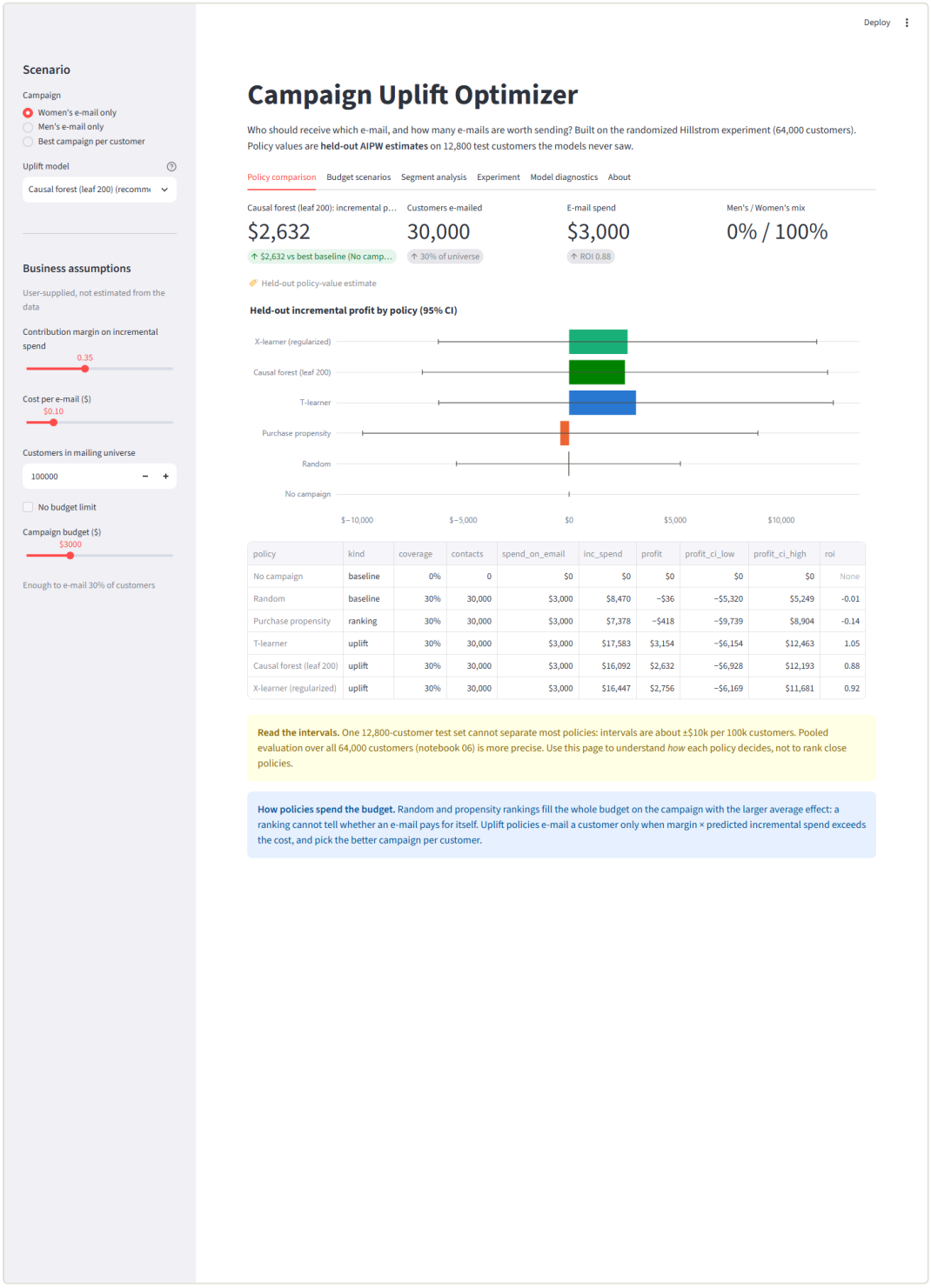
Held-out profit vs coverage. Dots mark the coverage each uplift model chooses by itself (break-even rule).

Sensitivity to economics



Margin × cost grid at the $3k budget (single test split, so read for direction). Blue = profit / advantage, red = loss. Diverging palette with a grey midpoint at $0.

14 Phase 5 · Interactive dashboard (app/dashboard.py, NB05)



policy	kind	coverage	contacts	spend_on_email	inc_spend	profit	profit_ci_low	profit_ci_high	roi
No campaign	baseline	0%	0	$0	$0	$0	$0	$0	None
Random	baseline	30%	30,000	$3,000	$8,470	−$36	−$5,320	$5,249	-0.01
Purchase propensity	ranking	30%	30,000	$3,000	$7,378	−$418	−$9,739	$8,904	-0.14
T-learner	uplift	30%	30,000	$3,000	$17,583	$3,154	−$6,154	$12,463	1.05
Causal forest (leaf 200)	uplift	30%	30,000	$3,000	$16,092	$2,632	−$6,928	$12,193	0.88
X-learner (regularized)	uplift	30%	30,000	$3,000	$16,447	$2,756	−$6,169	$11,681	0.92

Policy-comparison tab: headline metrics, profit with 95% CI per policy, policy table and evidence labels.

Controls and tabs

Sidebar input	Effect
Campaign: Women's / Men's / Best per customer	Which arms the allocation may use

Uplift model	Recommended = causal forest (leaf 200); also labels the pre-registered pick
Margin, cost per e-mail, universe size, budget / no limit	User-supplied assumptions, which re-run the allocation

Tabs: Policy comparison · Budget scenarios (profit-vs-budget curve plus margin × cost heatmap) · Segment analysis (observed vs predicted uplift, who is targeted) · Experiment (ATEs, randomization checks) · Model diagnostics (gain curves, AUGC, BLP, Qini) · About.

Design principles to explain

1. **No retraining in the UI.** Sliders re-run only the allocation on frozen test-set scores and re-value it with fixed AIPW scores, so no test outcome can leak into a model.
2. **One code path.** The app calls the same `scenario.run_scenario()` as notebook 04. NB05 asserts the maximum difference is $0.000000.
3. **Evidence labels on every number:** observed experimental estimate · held-out policy value · model prediction · user assumption · simulated scenario.
4. **Artifact contract:** the app receives no raw outcomes or treatment labels (asserted in NB05).
5. **Tested:** `streamlit.testing.v1.AppTest` runs the app headless, changes widgets, and checks that the headline metric updates without exceptions.
6. **Performance:** `@st.cache_data` caches scenarios; each allocation on 12,800 customers takes milliseconds.

15 Phase 6 · Final evaluation & reproducibility (NB06)

Repeated splits

The frozen pipeline was re-run on 6 different 80/20 splits (seeds 42, 7, 11, 23, 101, 2024). For the same policy, profit per 100k varies with an **SD of $2.6k–$7.6k across splits**, which shows that one split cannot rank close policies.

Pooled 5-fold cross-fitted evaluation: the primary result

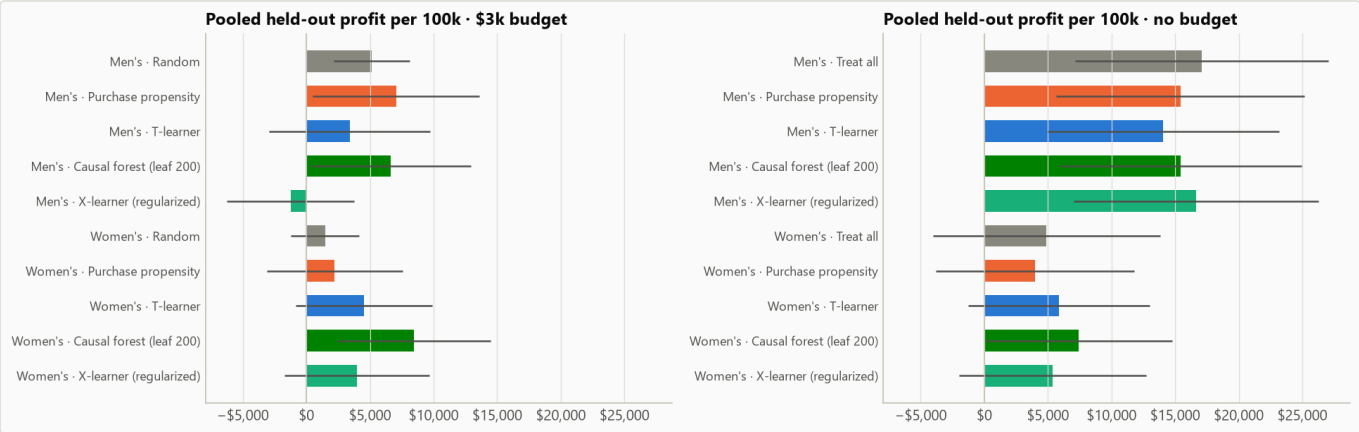
Every one of the 64,000 customers is used exactly once as held-out data. Frozen models train on 4/5 of the data, the policy is formed and valued on the remaining 1/5 (budget scaled), and the contributions are pooled. This roughly halves the standard error compared with one test split.

Profit per 100k (95% CI)	Random / Treat all	Propensity	Causal forest (leaf 200)	X-learner (reg.)
Women's, $3k budget	$1.5k	$2.2k	**$8.5k** (2.5k–14.5k)	$4.0k
Men's, $3k budget	$5.1k	$7.1k	$6.7k	−$1.2k
Women's, no budget	$4.9k	$4.0k	$7.4k	$5.4k
Men's, no budget	**$17.1k**	$15.4k	$15.4k	$16.7k

Statistically significant results (paired, pooled)

Women's, $3k budget: causal forest − random = **+$7,013** (95% CI $2,363 to $11,664; z = 2.96). Causal forest − propensity = **+$6,268** ($608 to $11,928; z = 2.17).

Men's, $3k budget: X-learner − random = **−$6,375** (z = −2.86): the pre-registered pick is significantly *worse* than random on Men's.



Pooled held-out profit by policy, with and without the budget.

The selection-metric lesson (post hoc, development data only)

Finalist	Dev AUGC (pre-registered rule)	Dev profit@30% Women's	Dev profit@30% Men's	Mean	Men's BLP
X-learner (regularized)	**0.073**	$9,071	$6,975	$8,023	0.004

Causal forest (leaf 200)	0.071	$7,929	$11,228	**$9,578**	0.311

A criterion matched to the decision (budgeted profit, per campaign, plus a positive-BLP gate) picks the causal forest **from development data alone**. The X-learner's Men's BLP of 0.004 was a red flag before the test set was opened. This analysis is labelled post hoc, and it is the most transferable lesson of the project.

Reproducibility record

- `reports/reproducibility.json`: data SHA-256, seeds, split fractions, fold count, Python and package versions, artifact sizes.
- `reports/results.json`: every headline number, written by code (nothing typed by hand).
- NB06 runs the full pytest suite (54 passed) from inside the notebook.

16 Results summary & business recommendations

Recommendations

1. **No budget cap: send the Men's e-mail to everyone** (≈ $17.1k contribution per 100k customers). Its effect is positive and roughly uniform, so no targeting rule beats it.
2. **Tight budget: target the Women's e-mail with the regularized causal forest** (≈ $8.5k per 100k at $3k, significantly better than random and propensity). A propensity-targeted Men's e-mail (≈ $7.1k) is comparable.
3. **Do not pick the campaign per customer with these models.** It is only as good as the *weaker* of the two effect models, and the Men's model is noise. An argmax over noisy CATEs is also biased toward the noisier arm.
4. **Validate prospectively** with a randomized holdout before rollout.

Methodological lessons (the "senior" talking points)

1. **Test for heterogeneity before modelling it.** NB01's Wald test predicted where uplift would pay off.
2. **Select on the decision metric.** Whole-curve AUGC averaged across campaigns hid a useless model.
3. **Calibration matters as much as ranking** when decisions use dollar thresholds (break-even, arm choice).
4. **Regularization beats algorithm choice** on noisy, rare outcomes.
5. **Plan for power.** A 20% test split had an MDE twice the effect size, and pooled cross-fitted evaluation fixed it.
6. **Report negative results:** T-learner, calibrated propensity, the AUGC selection rule and Men's targeting all failed, with the evidence.

17 Key design decisions (and the alternatives)

Decision	Chosen	Alternative	Why
Primary outcome	Spend ($)	Conversion, visit	Maps directly to profit: m·Δspend − c, with no assumed order value. Visit was used as a robustness KPI
Primary arm	Women's	Men's	Smaller ATE (more room for targeting to change the decision) and the only detectable spend heterogeneity
Propensities	Known (empirical arm shares)	Estimated e(x)	Randomized design, confirmed by the SRM test; makes AIPW unbiased by construction
Evaluator	Cross-fitted AIPW	Qini on raw outcomes, IPW	Never worse than IPW, handles three arms, valued on the full population. IPW and Qini were reported as checks
Validation	5-fold cross-fitting on 80% dev	Single 60/20/20 split	A single validation split's MDE was 2× the effect; cross-fitting halves the SE
Base learner	HistGradientBoosting, selected in NB07; RF baseline; XGBoost candidate	LightGBM, CatBoost, linear, neural nets	Chosen on nested-CV held-out policy value: tied on profit, fastest by 3–8×. See section 12
Meta-learners	Own implementation	CausalML	Transparent, about 150 lines, one shared interface; EconML only for the forest
Forest treatment model	DummyClassifier(prior)	Learned classifier	Assignment is random, so residualizing T only removes its known mean
Optimizer	Greedy + MILP check	LP / heuristic only	Greedy is provably optimal with equal costs; the MILP verifies it on real data
Results reporting	Per 100k customers	Per customer	Business-readable; the dashboard scales to any universe size
Dashboard	Streamlit on frozen artifacts	Live retraining	Prevents leakage, is fast, and every number is reproducible

18 Limitations & what you'd do next

Limitations

- **External validity:** one retailer, 2008, two-week window. Effects and costs today will differ.
- **Short horizon:** fatigue, unsubscribes and later purchases are unobserved; they are only proxied in the per-e-mail cost.
- **Noisy KPI:** 0.9% buyers, so close policies cannot be separated.
- **Assumed economics:** margin, cost and budget are inputs, and the dashboard shows the sensitivity.
- **Selection/evaluation overlap** in the pooled analysis. Any bias would favour the pre-registered pick, which still lost; comparisons against random and propensity are unaffected.
- **Multiple segments compared:** Holm correction is applied, and segments are framed as hypotheses.

Next steps

1. **Prospective test:** run the policy against a randomized holdout. Detecting a $5k per 100k profit difference at 80% power needs roughly 150–200k customers per arm.
2. **Pre-register a decision-matched selection rule:** budgeted profit per campaign plus a BLP gate.
3. **Shrink CATEs toward the ATE by their BLP slope** before allocating. This was identified post hoc and needs fresh data.
4. **Two-stage objective:** visit uplift × conversion given visit, to exploit the learnable visit signal.
5. **Richer features:** e-mail engagement history, category affinity, tenure.
6. **Policy learning:** optimize the policy directly (e.g. policy trees, Athey & Wager 2021) instead of plugging in CATEs.
7. **Long-term value:** add unsubscribe risk and 90-day spend as outcomes.

19 Numbers cheat sheet

Fact	Value
Customers / arms	64,000 · 21,307 M / 21,387 W / 21,306 control
Buyers / conversion rate	578 · 0.90%
Visit rate overall	14.7%
Control spend per customer	$0.65
SRM p / max |SMD| / assignment AUC	0.90 / 0.014 / 0.50
Men's ATE spend · conversion · visit	$0.77 · +0.68 pp · +7.66 pp
Women's ATE spend · conversion · visit	$0.42 · +0.31 pp · +4.52 pp
Relative spend lift M / W	+118% / +65%
Spend MDE on 20% split (W / M)	$0.82 / $0.91
Heterogeneity Wald p (spend) W / M	0.014 / 0.57
Split	80% dev (5 folds) / 20% test · seed 42
Features	12 pre-treatment
Economics (assumed)	35% margin · $0.10/e-mail · $3k per 100k (30% coverage)
Propensity AUC (W: treated / control)	0.573 / 0.529
Propensity AUC (M: treated / control)	0.579 / 0.621
Best dev mean AUGC	Propensity 0.076 · X-reg 0.073 · CF-200 0.071
BLP Women's: X default → regularized; CF 50 → 200	0.22 → 0.69; 0.34 → 0.79
Base-learner roles	RF baseline · XGBoost primary candidate · HistGB efficiency benchmark
T-learner held-out profit @ $3k (nested CV): RF / XGB / HistGB	$1.5k / $1.4k / **$2.2k** (tie)
T-learner no-budget profit, Women's: RF / XGB / HistGB	**$7.0k** / $3.7k / $6.3k
Fit time per T-learner: RF / XGB / HistGB	0.81 s / 0.29 s / 0.11 s
Default-settings comparison (NB03 §4c), mean AUGC HistGB / RF / XGB	+0.044 / +0.076 / +0.042
CF individual CIs including 0	≈ 90% (median width $2.23)
Visit AUGC W: uplift models vs propensity	0.0072–0.0076 vs 0.0030
Test-set Women's BLP: X-reg / CF-200	0.81 / 0.93
Greedy vs MILP	identical objective · 2 ms vs 13.3 s
Pooled W $3k: CF / propensity / random	$8.5k / $2.2k / $1.5k per 100k
CF − propensity (W, $3k), paired	+$6.3k (0.6k–11.9k), z = 2.17
CF − random (W, $3k), paired	+$7.0k (2.4k–11.7k), z = 2.96

Men's treat-all, no budget	$17.1k per 100k (best overall)
Split-to-split SD of profit	$2.6k–$7.6k
Tests	54 pytest + headless app test

20 How to run & demo

Setup (Windows PowerShell)

```
py -3.11 -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
# data: Data/Kevin_Hillstrom_MineThatData.csv (or data/raw/)
pytest                                      # 54 tests, ~30 s
jupyter nbconvert --to notebook --execute --inplace notebooks/0*.ipynb   # ~25 min
streamlit run app/dashboard.py
```

A 5-minute live demo script

1. **Experiment tab** (30 s): "Randomization is valid, and both e-mails work on average."
2. **Policy comparison**, Women's, $3k (1 min): "The uplift models contact only customers where margin × predicted uplift exceeds cost. Note the wide intervals, which is why I pooled across all 64k in notebook 06."
3. **Tick "No budget limit"** (30 s): "The model picks its own coverage, about 60%. A ranking can't do that."
4. **Switch to Men's** (30 s): "Here targeting does not help, so mail everyone. This was predicted by the heterogeneity test."
5. **Budget scenarios** (1 min): drag margin and cost, show the curve flattening and the heatmap.
6. **Segment analysis** (1 min): "New customers and women's buyers respond to the Women's e-mail; the policy targets them."
7. **Close** (30 s): "Nothing retrains here. Sliders re-run the optimizer on frozen held-out scores, and every number is labelled by evidence type."

Notebook tour (if asked to walk through code)

NB01 §5 (ATE table and forest plot) → NB02 §2 (AIPW sanity check) and §6 (propensity quintiles) → NB03 §2 (AUGC table), §3b (BLP) and §5c (calibrated-propensity failure) → NB04 §4 (profit vs coverage) and §7 (MILP) → NB06 §1c–1e (pooled results and selection lesson).

21 33 interview questions with answers

Grouped by theme. Each answer is written to be said aloud in about 45–90 seconds. "Follow-up" lines anticipate the next probe.

A · Project & business framing

Q1. Walk me through this project.

Use the 60-second pitch (Section 1): problem → randomized data → average effects → baselines vs uplift models → held-out policy evaluation → budget optimization → dashboard → key lesson.

Follow-up: "What was the hardest part?" Statistical power. Spend is rare and noisy, so the evaluation design (AIPW, cross-fitting, pooled evaluation) mattered more than model choice.

Q2. Why uplift modelling instead of a standard response / propensity model?

A response model predicts who buys. Profit comes from who buys *because of* the e-mail. Customers who would buy anyway score high on propensity but have zero incremental value, so contacting them wastes budget. Uplift models estimate $\tau(x) = E[Y(1) - Y(0) \mid x]$ directly, and the targeting rule becomes "contact if margin × $\tau(x)$ > cost".

Interestingly, in this data the propensity model was a decent uplift ranking for the Women's e-mail, because baseline demand was barely predictable. That was a campaign-specific accident that only the control group could reveal.

Q3. Why this dataset? What makes it suitable?

It is a **randomized** experiment with a no-e-mail control. Causal effects are therefore identified without assumptions about confounding, and the assignment probabilities are known, which makes doubly robust evaluation unbiased. It has two treatment arms, which allows campaign selection, and it is a well-known public benchmark. Its weakness is that it is old and spend is very rare, and the project treats that rarity as a central challenge.

Q4. What is the business impact, in one sentence?

Under a \$3k-per-100k budget, causal-forest targeting of the Women's e-mail earns about \$6.3k more contribution per 100k customers than propensity targeting (95% CI \$0.6k–\$11.9k). Without a budget cap, the evidence says to mail the Men's e-mail to everyone, worth about \$17k per 100k. The second insight saves money by *not* building targeting where it doesn't help.

B · Experimental design & statistics

Q5. How did you verify the randomization?

I used three checks. A sample-ratio-mismatch chi-square test against equal thirds gave $p = 0.90$. Standardized mean differences on all 12 features for both arms were all below 0.014, against a 0.1 threshold. As a joint test, a logistic regression could not predict assignment from the features (cross-validated AUC 0.50). If SRM had failed, I would have suspected a logging or assignment bug and would not have trusted any effect estimate.

Q6. Why use difference in means, and what does Lin regression adjustment add?

Under randomization, difference in means is unbiased and design-based, so it is the primary estimator. Lin (2013) adjustment regresses Y on T, centred covariates and their interactions with T, using HC2 robust errors. It is consistent and never asymptotically less precise. Here it reduced variance by under 3% because the features explain little of spend. I also ran a bootstrap and a permutation test because spend is heavy-tailed; all four estimators agreed.

Q7. Why Holm correction? Why not Bonferroni or Benjamini–Hochberg?

I tested six primary hypotheses (2 campaigns × 3 outcomes) plus many segment cells. Holm controls the family-wise error rate like Bonferroni but is uniformly more powerful. BH controls the false discovery rate, which suits exploratory screening of many segments. For confirmatory claims, FWER control is the stricter and more defensible choice.

Q8. What is the minimum detectable effect, and why did it matter?

MDE = $(z_{1-\alpha/2} + z_{power}) \times \sqrt{(\sigma_1^2/n_1 + \sigma_0^2/n_0)}$. For spend on a 20% test split, it was \$0.82, about twice the Women's ATE. So a single test split could not distinguish policies, which is exactly what happened in notebook 04. That motivated cross-fitting for model selection and a pooled 5-fold evaluation over all 64k customers for the final claims.

Q9. How did you know heterogeneity was worth modelling?

I ran a joint Wald test that all treatment × covariate interaction coefficients are zero. For Women's spend, $p = 0.014$, so there is heterogeneity. For Men's spend, $p = 0.57$, so there is none detectable. That cheap linear test predicted the final result: uplift targeting worked for Women's and failed for Men's. I'd run it first on any new campaign before investing in uplift models.

C · Causal ML models

Q10. Explain S-, T- and X-learners. When does the X-learner help?

S-learner: one model with treatment as a feature; $\tau = \mu(x,1) - \mu(x,0)$. It can shrink effects to zero if trees never split on T.

T-learner: separate models per arm; $\tau = \mu_1 - \mu_0$. Simple, but it differences two noisy estimates.

X-learner: fits μ_0 and μ_1, imputes individual effects using the *other* arm's model ($D_1 = Y - \mu_0(X)$ for treated, $D_0 = \mu_1(X) - Y$ for controls), regresses those on X, and blends with propensity weights. It helps when the signal is weak or the arms are imbalanced, because the second stage directly models the effect, which is usually smoother than the outcomes.

Q11. What is a causal forest and how does it give confidence intervals?

It is a random forest whose splits maximize heterogeneity in the *treatment effect* rather than outcome fit (Wager & Athey). **Honesty** means one subsample chooses the splits and a separate subsample estimates the leaf effects. That removes the adaptive-split bias and makes the estimates asymptotically normal, so CIs come from a variance estimate (bootstrap of little bags).

EconML's CausalForestDML also residualizes Y and T first (double machine learning). Because treatment here is randomized, I used a constant prior classifier for T. In this data, about 90% of individual CIs included zero.

Q12. What is the DR-learner?

Compute cross-fitted AIPW pseudo-outcomes Γ for every customer, whose conditional mean is τ(x) because the propensity is known, then regress Γ on X with any ML model. It is doubly robust and has good theoretical rates (Kennedy 2020). Here it was mid-pack (AUGC 0.047), because Γ for spend is extremely noisy, with an SD of about $34.

Q13. Why did the T-learner perform so badly?

Each arm's model fits a 0.9%-nonzero outcome, so each is noisy, and the difference of two noisy models has roughly double the variance with no shrinkage toward a common effect. Its predicted effects had an SD of $1.10 around a $0.46 mean, and a quarter of them were negative. The BLP slope was −0.09, meaning no held-out signal. Interestingly, its *segment averages* tracked observed segment effects very well (r = 0.98), so the noise is individual-level and averages out.

Q14. Why did a non-causal purchase-propensity model rank uplift well for the Women's e-mail?

It was trained on e-mailed customers, so it learns P(buy | e-mailed) = baseline + uplift. For Women's, baseline demand was barely predictable (AUC on control 0.53), so most of what it learned was responsiveness (AUC on treated 0.57). It became an implicit uplift model.

For Men's, it ranked control buyers better (0.62), so it learned "sure things", and its edge vanished. Lesson: whether propensity works for targeting is campaign-specific, and only a control group can tell you.

Q15. How did you tune hyperparameters, and what mattered most?

I used a small, pre-declared grid evaluated out of fold: default vs regularized variants (min_samples_leaf 200 → 600, max_leaf_nodes 15 → 8 for the X-learner; leaf 50 → 200 for the forest). Regularization was the single most important lever. It roughly tripled the BLP calibration slope (0.22 → 0.69 and 0.34 → 0.79) and made the forest the most seed-stable model (rank correlation 0.96). With rare, noisy outcomes, variance control beats algorithm sophistication.

Q16. Tell me about a hypothesis that failed.

I proposed a "calibrated propensity" model: keep the strong propensity ranking and convert it to dollars with two causal parameters, τ = a + b·p(x), fitted on out-of-fold doubly robust pseudo-outcomes. It should have inherited propensity's ranking. Instead, the two parameters were unstable across training folds: one fold collapsed to near-constant predictions, and in one Men's fold the slope turned negative, inverting the ranking. Even a 2-parameter fit is underpowered with 0.9% buyers. I reported it as rejected, with the per-fold diagnostic.

D · Evaluation & validation

Q17. How do you evaluate an uplift model when individual effects are never observed?

I evaluate at the *policy* level using the randomized data. Each customer gets an AIPW score Γ whose expectation equals their CATE. The value of e-mailing a set S is the average of Γ over S, with a CLT interval. From that come gain curves, AUGC, gain at 30% coverage and BLP calibration. I also report the classic Qini coefficient and an IPW cross-check. And I never describe a predicted CATE as a customer's "true" effect.

Q18. What does "doubly robust" mean, and why is it especially strong here?

AIPW combines an outcome model with inverse-propensity weighting. It is consistent if *either* the outcome model or the propensity model is correct. Here the propensity is known exactly from the design, so AIPW is unbiased no matter how wrong the outcome models are; they only affect variance. That is why I could use flexible models freely.

Q19. How did you prevent data leakage?

Five layers.

1. Only pre-treatment features, enforced by a unit test.
2. A 20% test set touched only once, after the model choice was frozen to a JSON file.
3. Selection via 5-fold cross-fitting on development data, so every score is out of fold.
4. AIPW nuisance models fitted within each evaluation fold, independent of the candidate models.
5. The dashboard gets no raw outcomes and never retrains.

Splits are stratified and seeded, and the campaign choice for baselines uses development ATEs, never test outcomes.

Q20. What is AUGC vs Qini, and what is the BLP slope?

AUGC is the area between the AIPW gain curve and the random-targeting line, in dollars per customer. **Qini** is the same idea computed on raw outcomes of a two-arm sample, with a scaling term N_T/N_C. Both measure *ranking*.

The **BLP slope** regresses Γ on the predicted CATE. A slope of 1 means predicted dollar differences are realized; 0 means noise. It measures *calibration*, which matters whenever decisions use thresholds like break-even or campaign comparison.

Q21. Your selected model lost on the final evaluation. What happened?

My pre-registered rule picked the highest AUGC averaged over the two campaigns. The X-learner won narrowly (0.073 vs 0.071) because of a very strong Women's ranking, while its Men's predictions were useless (BLP 0.004). In the pooled evaluation it was significantly worse than random on Men's ($z = -2.9$).

I then re-scored the finalists on development data only, using a decision-matched criterion: budgeted profit per campaign plus a calibration gate. That picks the causal forest. I labelled the analysis post hoc. The lesson is to select models on the metric of the decision they drive.

Q22. Why does the pooled 5-fold evaluation beat a single test split? Isn't it leaking?

Every customer is used exactly once as held-out data. Models are trained on the other 4 folds, and the policy is formed and valued within the held-out fold. Pooling all 64k roughly halves the standard error. The one caveat is that model *selection* used the 80% development data, which overlaps. That bias would favour the pre-registered pick, and the pick still lost, so the conclusion is conservative. Comparisons against random or propensity are unaffected.

Q23. How confident are you in the headline number?

Moderately. The causal forest vs propensity gain for Women's at $3k is +$6.3k per 100k, with a 95% CI of $0.6k–$11.9k (z = 2.17). It is significant, but the interval is wide. Profits for the same policy vary with an SD of $3–8k across random splits. So I'd present it as strong directional evidence that justifies a prospective test, not a guaranteed lift.

E · Optimization & decisions

Q24. How do you optimize targeting under a budget? Is greedy optimal?

The objective is to maximize $\Sigma\, a_{ik}(m \cdot \hat{\tau}_{ik} - c)$ subject to total cost ≤ B and at most one campaign per customer.

With equal cost per e-mail, the budget only caps the *count* of contacts, so the optimum is: pick each customer's best arm, drop negative net values, and take the top-N. That is exactly what greedy does.

I verified it with SciPy's HiGHS MILP on 25,600 binary variables: identical objective, 2 ms vs 13 s. With unequal costs it becomes a knapsack problem; I'd then use the MILP or a Lagrangian approach.

Q25. Why didn't "best campaign per customer" win?

Choosing an arm per customer needs a reliable comparison of two CATEs. The Men's CATE model carried no signal, so the argmax was largely noise. Worse, an argmax over noisy estimates is biased toward the noisier arm (the winner's curse). A per-customer choice is only as good as the weaker effect model. The remedy is to shrink each CATE toward its ATE using its BLP slope; with a slope near 0, that defaults to the better-average campaign.

Q26. How did you choose the economics, and how sensitive are the conclusions?

Margin (35%), cost ($0.10 per e-mail, including a fatigue allowance) and budget ($3k per 100k) are explicit user-supplied assumptions in one config file. Notebook 04 shows a margin × cost grid, the dashboard lets anyone change them, and every number is labelled as an assumption or a simulated scenario. The direction of the conclusions holds across most of the grid. Only at high contact cost ($0.30) does targeting's advantage flip, because few contacts pay off.

Q27. Why spend as the primary outcome, not conversion?

Spend is in dollars, so incremental profit is simply margin × incremental spend − cost, with no assumed order value and no double counting. Conversion would need a value-per-conversion assumption. The cost is noise: spend is heavy-tailed and rare. So I used visit as a higher-signal robustness KPI, which showed the models work well when the signal is there.

F · Engineering, deployment & next steps

Q28. How is the project engineered for reproducibility?

All logic is in an importable `src/` package, and notebooks only explain and call it. Everything is driven by one config file with fixed seeds. Pinned requirements, a data SHA-256 and package versions go into `reproducibility.json`, and headline numbers are written by code to `results.json`. There are 54 pytest tests: data contract, leakage, estimators on synthetic data with known truth, budget never exceeded, greedy = MILP, profit accounting. All six notebooks re-execute from scratch with nbconvert, and the dashboard is tested headless.

Q29. How would you deploy this and prove it works in production?

Score customers offline before each send with the frozen model, then run a randomized test: e.g. 80% policy, 10% random, 10% no e-mail. Measure incremental profit per arm with a pre-registered analysis. Detecting a $5k per 100k difference at 80% power needs about 150–200k customers per arm, so plan duration accordingly. Monitor drift in feature distributions and BLP calibration, refit on fresh randomized data, and track long-run outcomes such as unsubscribes. The dashboard becomes the planning tool for budget scenarios.

Q30. What would you do with more time or data?

1. Pre-register a decision-matched selection rule.
2. BLP-based shrinkage of CATEs before allocation.
3. Direct policy learning (policy trees) instead of plug-in CATEs.
4. A two-stage visit → purchase model to exploit the learnable visit signal.
5. Richer features such as engagement history and category affinity.
6. Long-term outcomes: 90-day value and unsubscribe risk.
7. A fresh randomized holdout to confirm the Women's targeting gain.

Q31. Can you use Random Forest or XGBoost inside a T-learner? Which base learner did you choose, and how?

Yes. T-, S-, X- and DR-learners are *meta*-learners, so any regressor can play $\hat{\mu}$. My code takes `base="rf" | "xgb" | "hgb"`, using scikit-learn's RandomForestRegressor and HistGradientBoostingRegressor directly, and XGBoost through its scikit-learn API.

I gave each a role up front: **Random Forest as the baseline, XGBoost as the primary candidate, HistGradientBoosting as the efficiency benchmark**. I then chose the winner on *held-out causal-policy value*, not reputation. The setup was the same features and splits, an equal 6-config tuning budget scored on budgeted profit, and nested cross-validation over all 64,000 customers, so every customer was held out exactly once.

HistGradientBoosting won: $2.2k per 100k vs $1.5k (RF) and $1.4k (XGBoost) at the $3k budget. That is a statistical tie (HistGB − RF +$0.7k, 95% CI −$2.4k to +$3.9k), but it is 2.7× faster than XGBoost and 7.5× faster than RF to fit. With no budget, RF won because its effect sizes were better calibrated.

Follow-up: "So which is better?" For this outcome they rank people about equally well, so I chose the fastest and most stable one, and I'd switch to RF for threshold-based (no-budget) policies.

Q32. XGBoost is usually the strongest tabular model. Why did it lose here?

Because a T-learner *subtracts* two outcome models, and 99.1% of outcomes are $0. Any noise each model learns survives the subtraction. Boosting adds trees that fit whatever the previous trees missed, and on this outcome most of that is noise from a few large purchases.

Without early stopping, XGBoost's dollar predictions were uncalibrated (Women's BLP ≈ 0). The budgeted policy only needs a ranking, so it tied there. But the no-budget policy needs correct effect *sizes* to decide who clears break-even, and there it lost $3–5k per 100k vs RF ($z \approx -1.8$ to -1.9). The base learner that fits the least noise wins: averaged forests or early-stopped HistGB.

Follow-up: "How would you make XGBoost competitive?" Use early stopping on a validation fold, a smaller learning rate with more rounds, a larger λ, and possibly a Tweedie objective for zero-inflated spend. Then re-test on fresh held-out data rather than the data already used to compare.

Q33. How did you compare efficiency, and which model is most efficient?

I timed every T-learner fit in the study (250 per base learner) on the same machine, with each library using all CPU cores. HistGB fit in 0.11 s per T-learner, vs 0.29 s for XGBoost and 0.81 s for Random Forest. Prediction was 0.010 s vs 0.015 s vs 0.176 s. Random Forest is slowest because it builds and then walks 300 deep trees. HistGB is fastest because it bins features into ≤ 255 histogram buckets and stops early.

Since policy value was statistically tied, HistGB gives the same held-out profit for a fraction of the compute. That matters for scoring millions of customers before each send, or for retraining often.

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22 Glossary

Term	Meaning
ATE / CATE / uplift	Average effect / effect conditional on features / CATE used for targeting
Potential outcomes	Y(1), Y(0): outcomes with and without treatment; only one is observed
Randomized controlled trial	Treatment assigned by chance, so groups are comparable and effects are identified
SRM	Sample-ratio mismatch: arm sizes inconsistent with the design, a sign of bugs
SMD	Standardized mean difference: balance measure; <0.1 is negligible
Welch interval	CI for a difference in means allowing unequal variances
Lin adjustment	Regression with centred covariates and T × X interactions, for precision
HC2	Heteroskedasticity-robust standard errors
Holm correction	Step-down multiple-testing control of the family-wise error rate
MDE	Smallest effect detectable with given power and α
IPW	Inverse propensity weighting: reweights observed outcomes by 1/P(treatment)
AIPW / doubly robust	IPW plus an outcome-model correction; consistent if either model is right
Cross-fitting	Fit on some folds and predict on held-out folds, so every prediction is out of sample
Meta-learner	Builds a CATE estimator from standard ML models (S/T/X/DR)
Honest forest	Separate samples for choosing splits and estimating leaves, giving valid CIs
DML	Double machine learning: residualize Y and T on X, then estimate effects
Gain curve / AUGC	Cumulative incremental outcome vs share targeted / area above random
Qini	Classic uplift curve on raw two-arm outcomes
BLP slope	Calibration of predicted CATEs on held-out data (1 = well scaled)
Policy / policy value	Rule from features to action / expected outcome if the rule were applied
Paired comparison	Difference of two policies on the same customers, giving a tighter CI
Break-even rule	Contact only if margin × predicted uplift exceeds cost
MILP	Mixed-integer linear program, solved exactly (HiGHS) to verify greedy
Winner's curse	Choosing the max of noisy estimates overstates it and favours noise
Pre-registration	Fixing the analysis and selection rule before seeing the evaluation data

23 Interview tips & traps to avoid

Do

- **Lead with the decision, then the method:** "Mail Men's broadly; target Women's with a causal forest under a budget."
- **Quote numbers with intervals** ("+$6.3k, 95% CI $0.6k–$11.9k") and say what they're "per" (100k customers).
- **Own the negative results.** They show judgement: T-learner noise, calibrated propensity, AUGC selection, Men's targeting.
- **Connect to product work:** heterogeneity tests before building, decision-matched metrics, power planning, holdout-based rollout.
- **Draw the pipeline** (Section 6) on a whiteboard if asked about architecture.

Traps

- Saying a customer's predicted CATE is their "true effect". It is a model estimate; individual effects are unobservable.
- Confusing *purchase propensity* with the *propensity score* e(x).
- Claiming uplift "always beats" propensity. Here it depended on heterogeneity existing.
- Quoting the single-test-split profit table as the final answer. The pooled evaluation is the primary result.
- Calling the post-hoc selection analysis "out-of-sample validation". It is labelled post hoc for a reason.
- Saying "the model lifts revenue 118%". That is the relative ATE of the Men's e-mail vs no e-mail, not a model effect.
- Forgetting the economics are assumptions (35% margin, $0.10 cost).

Likely rapid-fire checks

If asked...	Answer
Assignment probability?	≈ 1/3 per arm, known by design; empirical shares used
How many features?	12 pre-treatment (8 raw columns, history_segment dropped)
Base learner?	HistGB (winner of the nested-CV held-out study, fastest); RF = baseline; XGBoost = primary candidate (lost)
Forest library?	EconML CausalForestDML, 400 trees, honest, min leaf 200
Test set size?	12,800 customers (20%), touched once
Why 5 folds?	Balances training size (80% per fit) with evaluation coverage
Runtime?	~1 min per campaign for 7 models × 5 folds; full notebooks ≈ 25 min